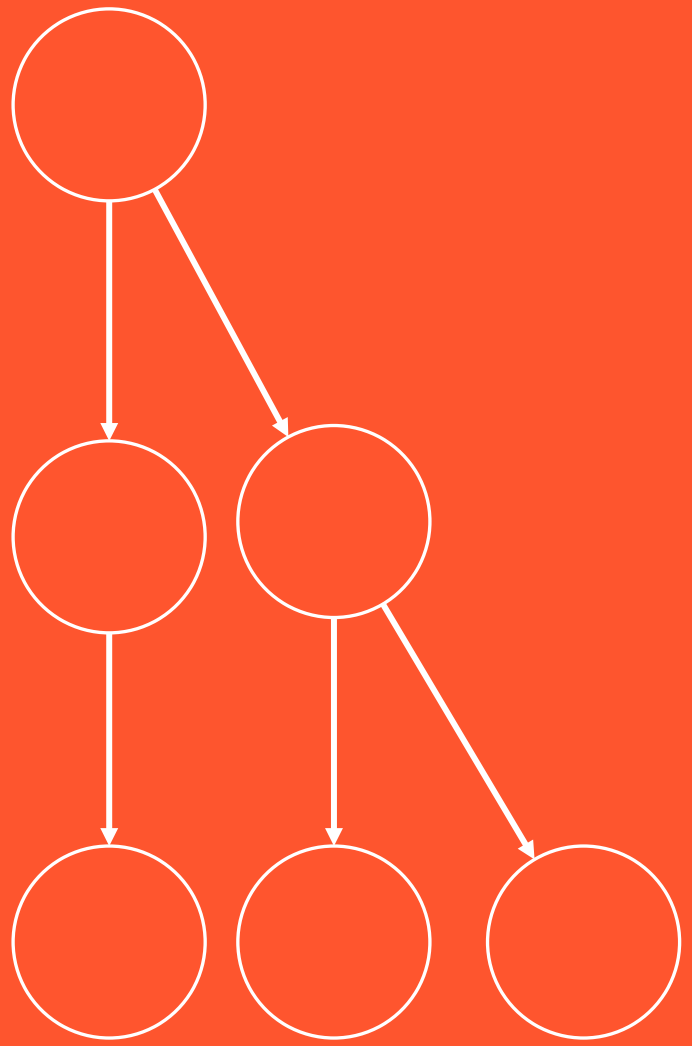


BCOG 100

Perception and Action

Lecture 8.3 — Receptor Sheets, Maps, and the Cerebellum

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

To Start: What Would It Take?

In your group, three minutes.

Every procedure in the second lecture could run in any substrate. Pick one: the delay line, a copy of the command, or learning a mapping from mismatch.

What would a nervous system need, to run it? What kind of cells, wired how, and how fast?

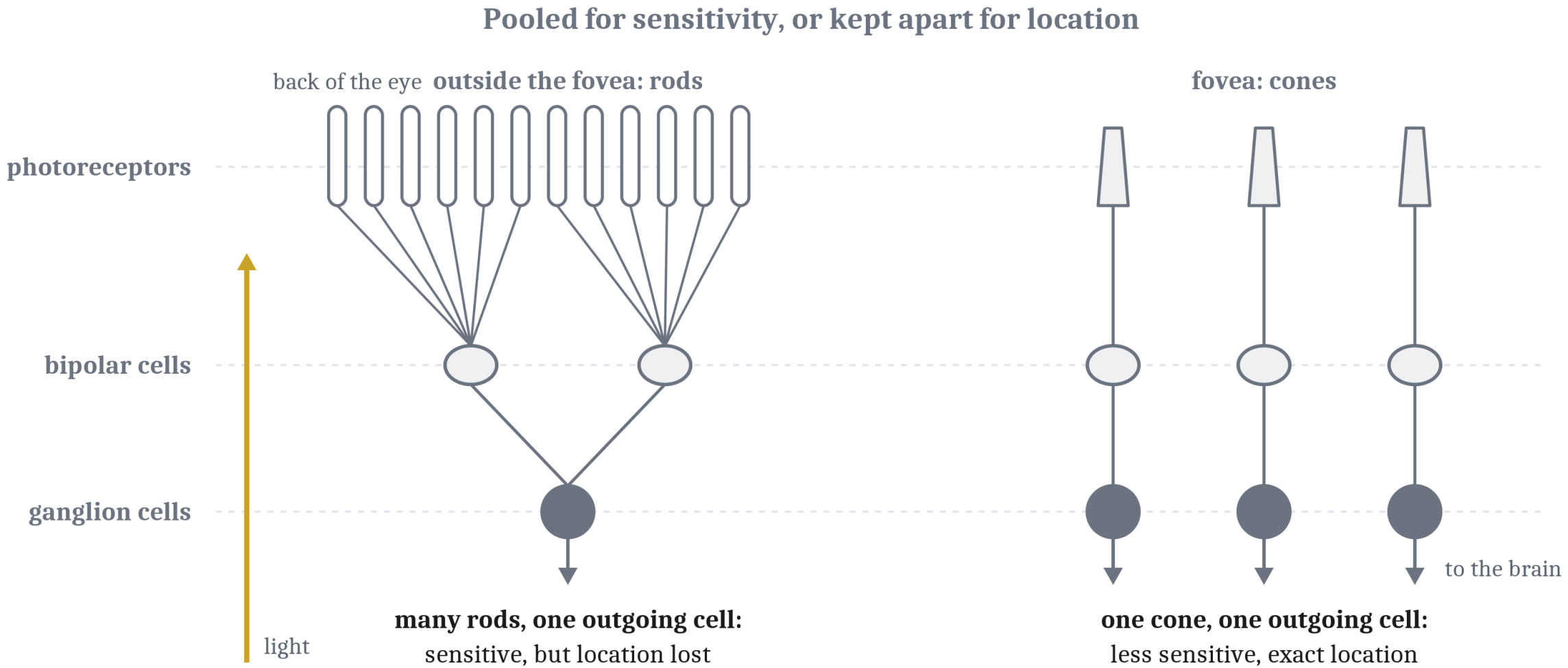
From Receptor Sheet to Cortex

Most receptor sheets keep their layout as maps: **retinotopy**, **tonotopy**, **somatotopy**. Of the senses that reach the cortex, all but smell pass through the thalamus.

Sense	Receptors laid out by	What the layout gives free	Route to the cortex
Vision	Direction in the visual field	Direction, relative to the eye	Thalamus (lateral geniculate nucleus), then primary visual cortex
Hearing	Frequency	Pitch, not direction	Brainstem stations, thalamus, auditory cortex
Balance	Axis of rotation, direction of tilt	Which way the head turns or tilts	Brainstem and cerebellum; thalamus to several areas, no single primary area
Touch and proprioception	Location on the body	Where on the body	Spinal cord and brainstem, thalamus, somatosensory cortex
Smell	Scattered, then sorted by receptor type	Nothing about location	Olfactory bulb straight to olfactory cortex
Taste	Across the tongue, every region sensing every taste	Which of five basic tastes	Brainstem, thalamus, taste cortex
Lateral line and electric sense (fishes)	Along the body surface	Where along the body	Cerebellum-like structures in the hindbrain, then the midbrain

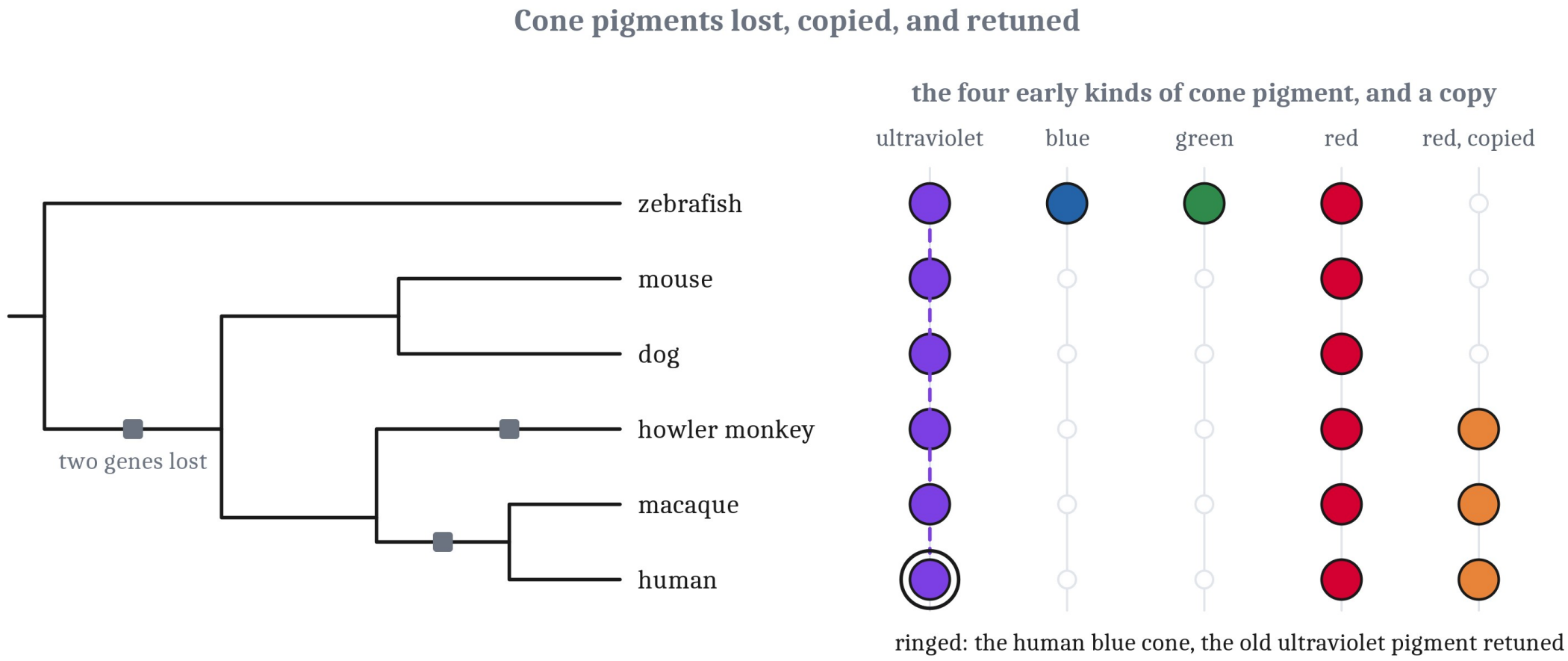
The Retina Computes

A **photoreceptor** is the retina's receptor: a **rod** for dim light, a **cone** for daylight, detail and color. Wiring trades sensitivity for location.



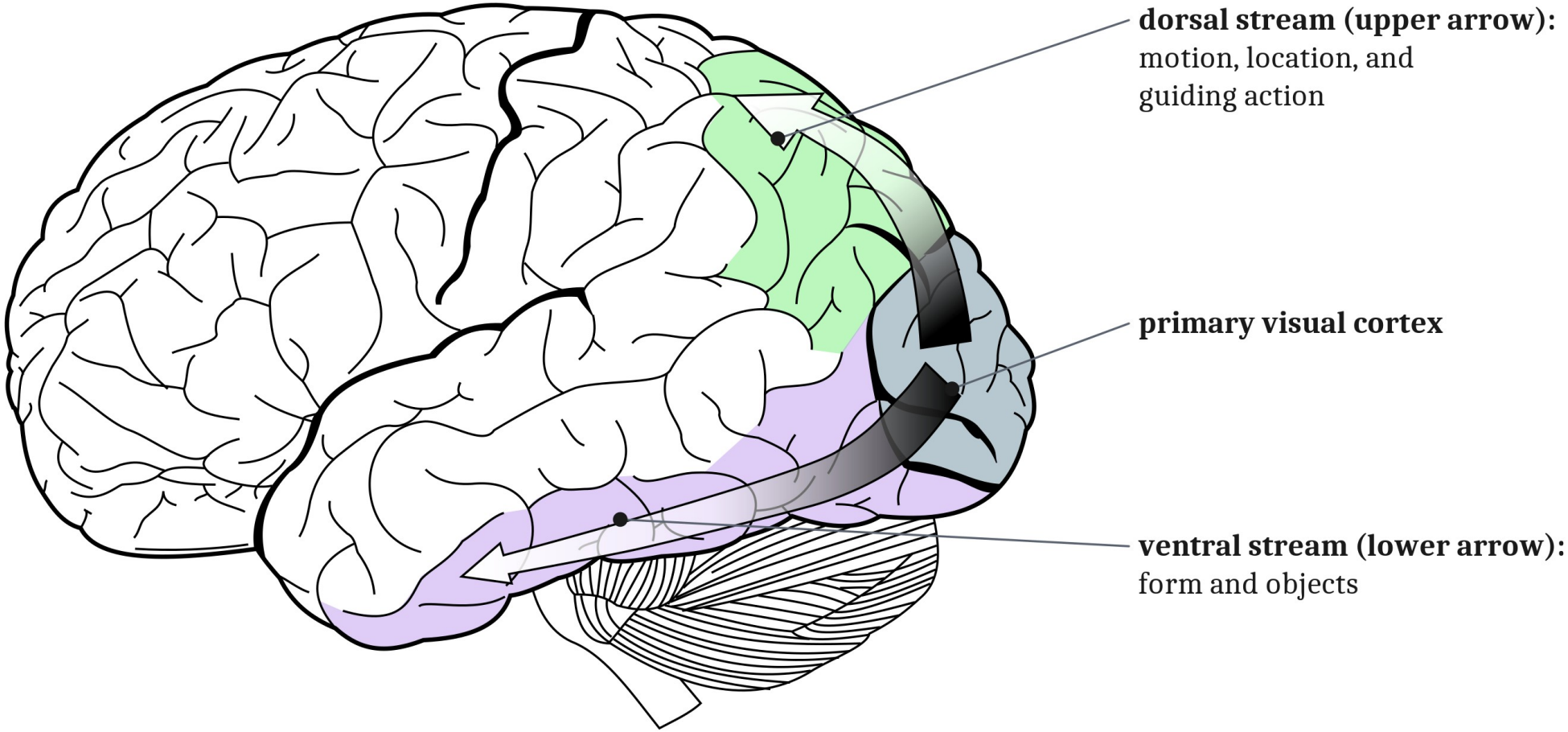
Cone Classes, Lost and Duplicated

Early vertebrates had four cone classes; mammals lost two. A third cone in our ancestors came from a duplicated gene, and the human blue cone carries the old ultraviolet pigment.



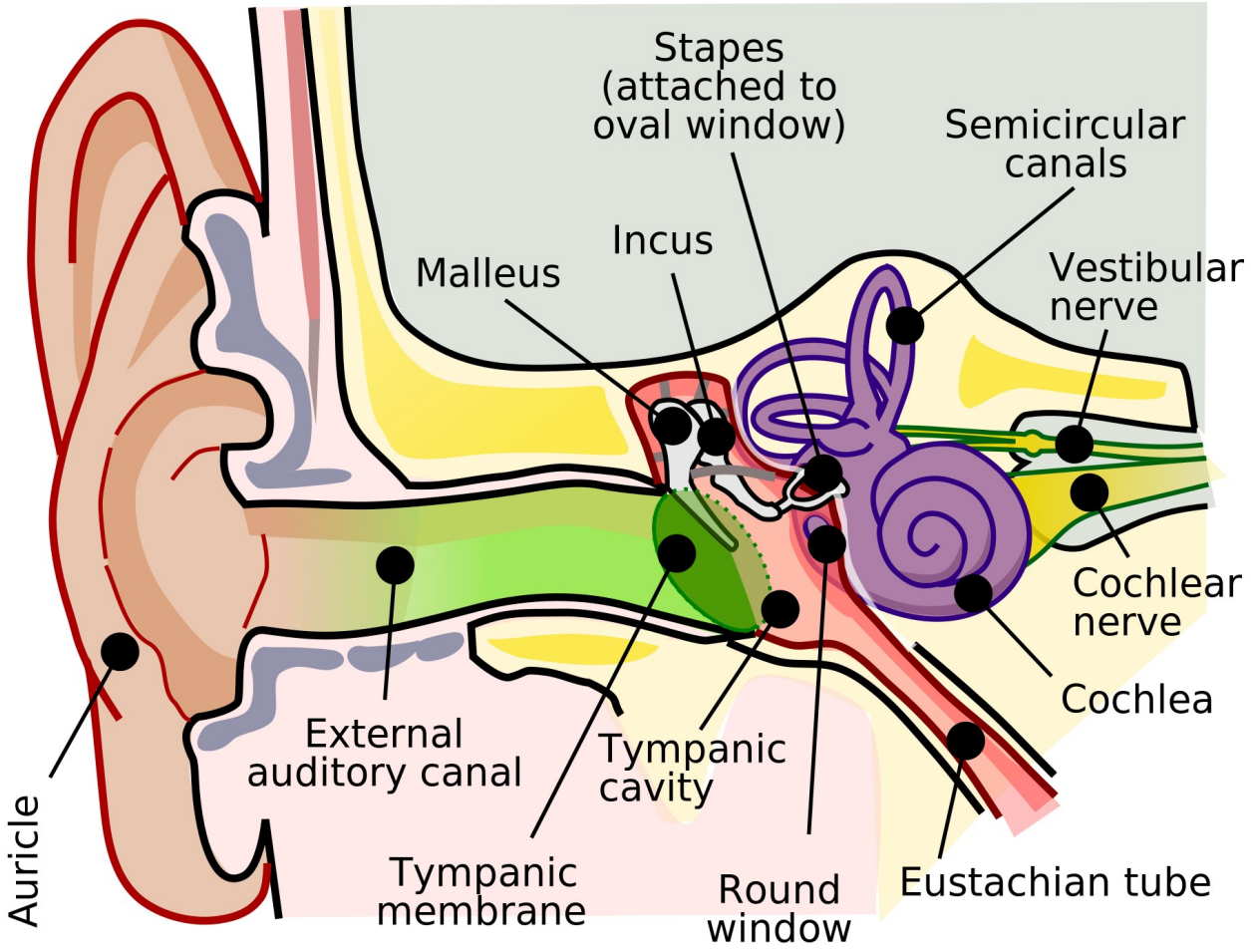
Two Streams, Specialized

The ventral stream for form and objects; the **dorsal stream** for motion, location and guiding action. Specialized, not independent.



The Ear, from Old Parts

The middle ear's bones began as a jaw brace and a jaw joint. An eardrum concentrates the force of sound in air, and eardrums evolved late, separately, at least five times.



Its labels, in this chapter's words

Auricle
outer ear

Tympanic membrane
eardrum

Tympanic cavity
middle ear

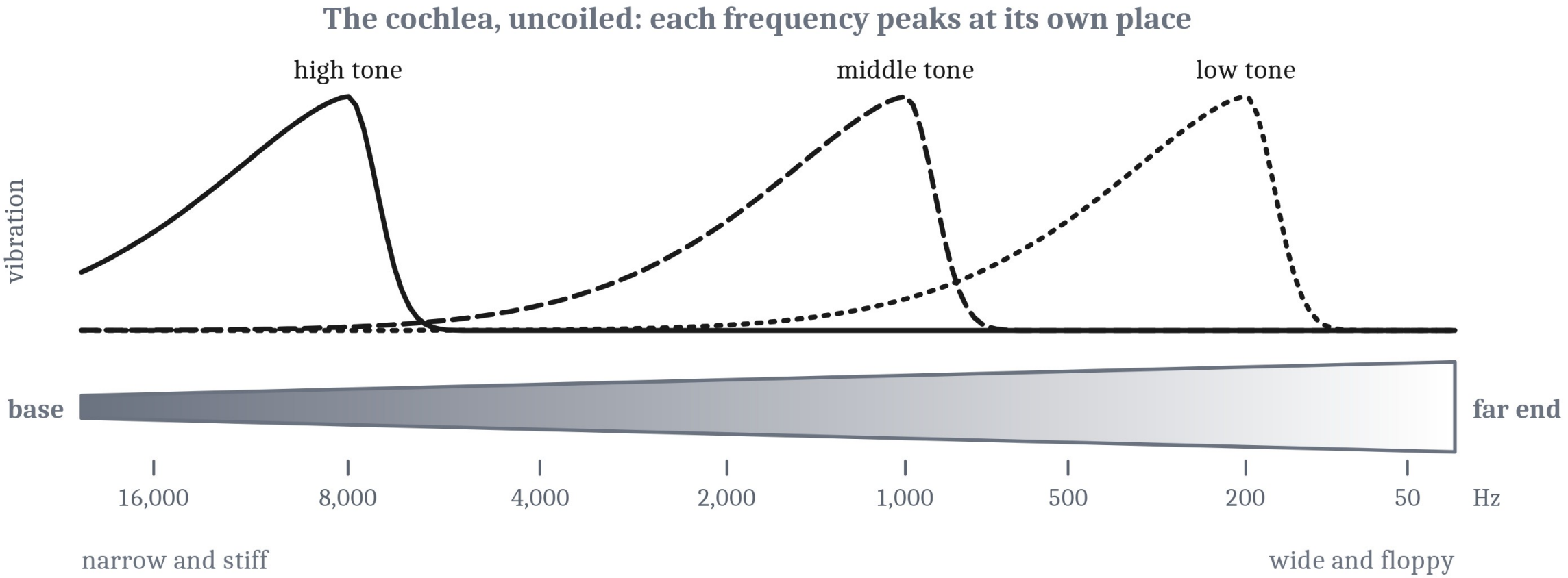
Malleus, incus, stapes
the three bones

Eustachian tube
tube to the throat

Semicircular canals
balance organs

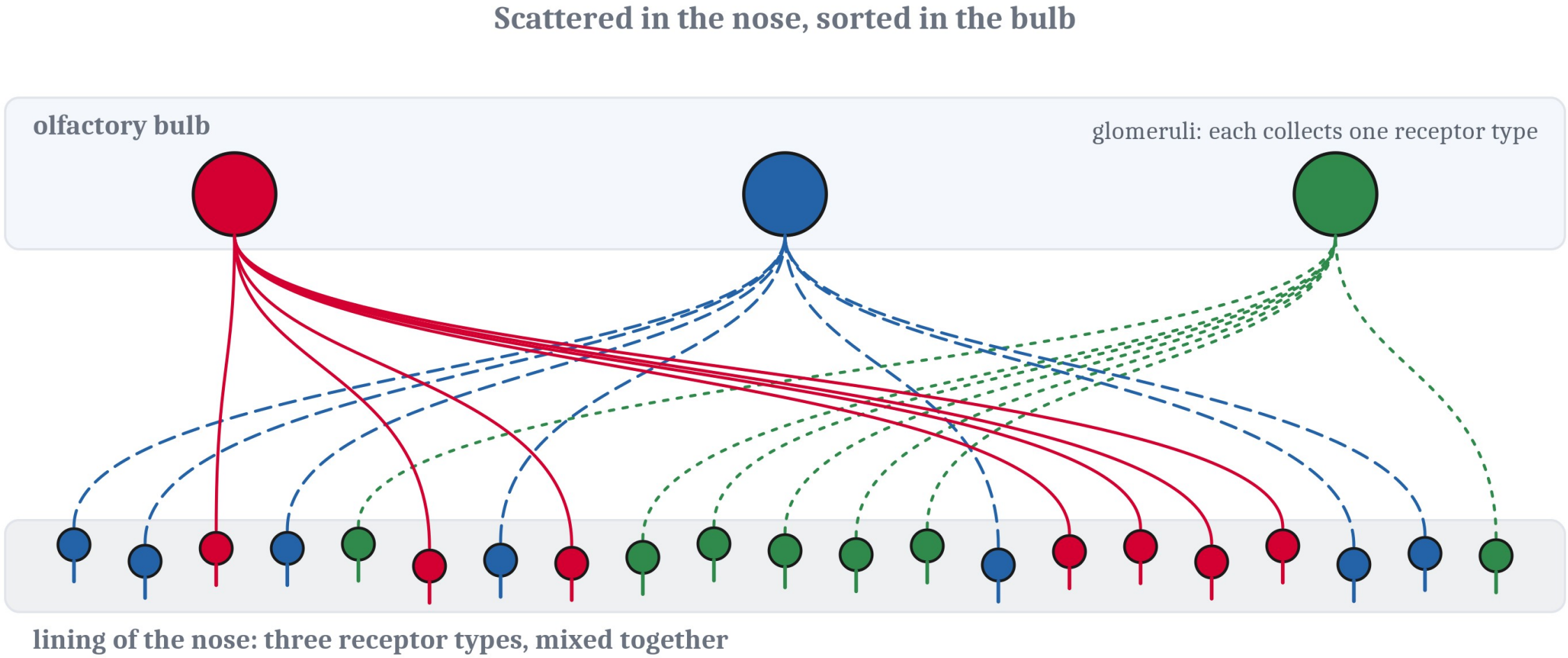
The Cochlea, the Hair Cell, and the Owl

The cochlea is a bank of filters by pitch. The **hair cell** opens its channels directly, fast enough for timing. The **superior colliculus** aligns auditory space with visual space.



Smell and Touch

Smell neurons of one receptor type, scattered across the nose, converge on the same few glomeruli. A **glomerulus** is a small cluster of connections in the olfactory bulb. Touch keeps location, mapped as a **homunculus**.



Motor Cortex Is Not a Mirror

Motor cortex maps the body coarsely, in overlapping zones organized around what the body does. Direction is a vote across many neurons.

Two maps of the body, side by side in the cortex



somatosensory map
each part in its own segment

motor map
the same order, in overlapping zones

each strip is a slice of cortex, running from the top of the head down the side

Image Sources (1 of 2)

Slide	Image	File	License	Source
4	Rods converge, cones do not	fig-retina-convergence.png	CC BY 4.0	Jon Willits — drawn for the Module 8 reading · own work
5	Cone pigments across the vertebrates	fig-cone-classes.png	CC BY 4.0	Jon Willits — drawn for the Module 8 reading · own work
6	The two visual streams	fig-two-streams.png	CC BY-SA 3.0	Selket, adapted from Gray's Anatomy; labels added for the Module 8 reading · https://commons.wikimedia.org/wiki/File:Ventral-dorsal_streams.svg
7	The human ear	fig-ear.png	CC BY 2.5	Lars Chittka and Axel Brockmann, vectorized by Inductiveload; labels removed and a key added for the Module 8 reading · https://commons.wikimedia.org/wiki/File:Anatomy_of_the_Human_Ear.svg
8	The cochlea's map of pitch	fig-cochlea-tonotopy.png	CC BY 4.0	Jon Willits — drawn for the Module 8 reading · own work
9	Scattered receptor neurons, sorted into glomeruli	fig-glomeruli.png	CC BY 4.0	Jon Willits — drawn for the Module 8 reading · own work
10	The sensory map and the motor map	fig-two-homunculi.png	CC BY 4.0	Jon Willits — drawn for the Module 8 reading · own work
11	The vestibulo-ocular reflex	fig-vor.png	CC BY 4.0	Jon Willits — drawn for the Module 8 reading · own work

Image Sources (2 of 2)

Slide	Image	File	License	Source
12	A learned negative image	fig-negative-image.png	CC BY 4.0	Jon Willits — drawn for the Module 8 reading, schematic after Bell (1981) · own work
13	Three jobs every sense does	fig-every-sense.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work